

Lesson 3 Logical Equivalence

1. Use a truth table to prove the following logical equivalences.
 - a) $(p \wedge q) \wedge r = p \wedge (q \wedge r)$ Associative
 - b) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ Distributive
 - c) $p \rightarrow q = \sim q \rightarrow \sim p$ Contrapositive
 - d) $p \vee (q \wedge p) = p$ Absorption
 - e) $\sim(p \leftrightarrow q) = p \underline{\vee} q$ Exclusive OR
2. Use a truth table to prove that the following statements are tautologies.
 - a) $[(p \vee q) \wedge \sim p] \rightarrow q$
 - b) $[(p \rightarrow \sim q) \wedge q] \rightarrow \sim p$
 - c) $(p \wedge q) \rightarrow p$
 - d) $p \rightarrow (p \vee q)$
 - e) $(p \wedge q) \rightarrow (p \vee q)$
3. For each of the following conditionals, write it using “or” (without “if...then”)
 - a) If p is a prime number, then $\sqrt{p}$ is not rational.
 - b) If x is prime, then x is odd or $x = 2$.
4. Write the following statements as a conditional statement, “If... then...”
 - a) An integer is odd or even.
 - b) x is not even or x^2 is even.
5. Write the converse, inverse, and contrapositive, of the following statements.
 - a) If x is prime, then x is odd or $x = 2$.
 - b) If x and y are rational numbers, then xy is rational.
 - c) If $x = -1$ or $x = 1$, then $x^3 = x$.
6. Write a negation for each of the following statements.
 - a) Both x and y are even.
 - b) AB is a symmetric matrix but BA is not.
 - c) $\sqrt{x}$ is an integer or irrational.
 - d) xy is even, or x and y are odd.
 - e) $x > 1$, and $y = -\sqrt{x}$ or $y = \sqrt{x}$
 - f) If p is a prime number, then $\sqrt{p}$ is not rational.
 - g) If x is prime, then x is odd or $x = 2$.
7. Use the basic equivalence laws to prove that $\sim(q \rightarrow \sim p) = p \wedge q$.
8. Simplify $(p \wedge \sim q) \rightarrow (p \wedge q)$ to $\sim p \vee q$ using the basic equivalence laws.
9. Let p be “ x is odd” and q be “ x is even”, and
 - u = “ x is odd if and only if x is not even.”
 - v = “ x odd or even but not both.”Write u and v in terms of p and q , and use a truth table to show that $u = v$.

10. Determine if the following equivalences are true. Give a proof.

- a) $\sim(\sim p \wedge \sim q) = p \vee q$
- b) $p \vee (q \wedge r) = (p \vee q) \wedge r$
- c) $p \vee (q \wedge r) = (r \wedge q) \vee p$
- d) $\sim(p \rightarrow q) = \sim p \rightarrow \sim q$
- e) $\sim(p \rightarrow \sim q) = p \wedge q$
- f) $p \leftrightarrow q = \sim p \leftrightarrow \sim q$
- g) $p \rightarrow \sim p = p \wedge \sim p$
- h) $p \wedge (p \rightarrow q) = p \wedge q$
- i) $p \vee (p \rightarrow q) = p \vee q$
- j) $\sim p \rightarrow (p \wedge q) = p$
- k) $p \rightarrow (p \wedge q) = T$
- l) $p \rightarrow p = T$
- m) $p \rightarrow \sim p = F$